BIOCHEMISTRY

Q-1: Define and classify carbohydrates with examples. Why is sucrose a non-reducing sugar?
Q-2: Mention the importance of glucose. Name the important dietary sources of carbohydrates.
Q-3: Discuss 5 polysaccharides with their biomedical importance.
Q-4: Write three diseases associated with carbohydrate metabolism.
Q-5: Write short notes about the Benedict test with interpretation.
Q-6: Define and classify diabetes mellitus. Give the difference between type-I and type-II diabetes mellitus.
Q-7: Write down the clinical features of Diabetes mellitus. Discuss the lab diagnosis of DM.
Q-8: Define and classify lipids. Give the difference between fat and oil.
Q-9: What is neutral fat?
Q-10: Define and classify fatty acid. Name the sources of fatty acid.
Q-11: Define and classify lipoprotein. Mention the constituents and function of lipoprotein. Give the origin of different types of lipoprotein.
Q-12: Why is LDL cholesterol bad and HDL cholesterol good for our health?
Q-13: Give the dietary source of cholesterol. Write down the function & biomedical importance of cholesterol.
Q-14: Define Catalyst, enzyme, and co-enzyme. Give the properties and clinical importance of enzymes.
Q-15: Give the IUBMB classification of enzymes with examples. Mention the clinical importance of CK-MB and ALT. Describe the factors affecting enzyme activity.
Q-16: Short note on functional and non-functional plasma enzymes.
Q-17: Short note on lipid profile.
Q-18: Write down the functions of prostaglandins.
Q-19: Define and classify hormones with examples. Name an organ with endocrine function. Name the hormones of the pancreas.
Q-20: Short note on dialysis.
Q-21: Write a short note on isoenzymes.
Q-22: Write down the functional classification of protein. Write about denaturation of protein. What is an essential amino acid with an example? What is a peptide bond?
Q-23: Define acid, base, pH, and pH scale with an example. Draw the pH scale. Write four important points about pH. How is the normal pH of the body maintained at a constant level? Name the methods of determination of pH.

Q-24: Define and classify buffers. Name some important blood buffers. Which one is the most important in blood and why? Name some important colloids in blood.
Q-25: Define standard solution with an example. Classify solutions on the basis of tonicity with an example. Write down the advantages of crystalloid solutions.
Q-26: Define radioisotope with an example. Give the biomedical importance of isotopes in medical science. Mention 3 advantages and disadvantages of radioactive isotopes.
Q-27: Define nucleic acid, DNA, RNA, transcription, and central dogma. Give the difference between DNA and RNA.
Q-28: Enumerate different types of RNA with their functions. Mention the salient features of the structure and function of DNA.
Q-29: Define body fluid. Give the distribution of total body fluid in different compartments in a 70kg adult male.
Q-30: Write down the acid-base parameters with their normal range. Classify acid-base disorders with their primary defect. What is the plasma anion gap?
Q-31: Name the samples used for biochemical analysis/clinical biochemistry lab. Write about different types of laboratory hazards.
Q-32: What is quality control of a laboratory and how can it be maintained?
Q-33: What are the differences between serum and plasma? Name some biochemical tests which are done by fresh plasma. What are the emergency tests that we have done in the laboratory?
Q-34: Name the common analytical processes used in the clinical lab. Write the principle of a photometer. What are the differences between photometry and electrophoresis?
Q-35: Name the four major components of gastric juice with each function. Why is normal gastric juice secretion essential for erythropoiesis? Name the protein-splitting enzymes of the GIT.
Q-36: What are the biochemical changes occurring in collected blood when examination is delayed? How is hemolysis of blood prevented? Write down the preparation to be taken during sample collection?
Q-37: Write down a normal daily water intake and output chart.
Q-38: Define recombinant DNA technology. Write down the application of recombinant DNA technology.
Q-39: Short note on PCR.
Q-40: Short note on thyroid function tests.
Q-41: Name the thyroid diseases and tests related to them.
Q-42: Justify that bile is a digestive juice. Describe the steps of carbohydrate digestion and absorption.
Q-43: Write a short note on Anticoagulants.
Q-44: Short note on SI units.

Q-45: Short note on serum electrolytes.
Q-46: Short note on cardiac markers.

Q1. Define and classify carbohydrates with examples. Why is sucrose a non-reducing sugar?

Carbohydrate: Carbohydrates are aldehyde (-CHO) or ketone (-C=O) derivatives of higher polyhydric alcohols, or the compounds which produce these derivatives on hydrolysis.

Classification of Carbohydrate:

1. **Monosaccharide:** The simplest form of carbohydrate which cannot be hydrolyzed further into simpler forms. They contain 3 to 9 carbon atoms.
    - e.g., Glucose, Fructose, Galactose, etc.
2. **Disaccharide:** A carbohydrate produced by the condensation of two monosaccharide units joined by a glycosidic bond.
    - e.g.,
        - **Maltose** - Consists of two glucose molecules.
        - **Lactose** - Consists of one glucose and one galactose molecule.
        - **Sucrose** - Consists of one glucose and one fructose molecule.
3. **Oligosaccharide:** A carbohydrate produced by the condensation of 3 to 10 monosaccharide units joined by glycosidic bonds.
    - e.g., Maltotriose, Raffinose.
4. **Polysaccharide:** A polymer of monosaccharides composed of more than 10 monosaccharide units connected by glycosidic bonds.
    - e.g., Starch, Glycogen, Cellulose, Mucopolysaccharide, Glycoprotein, etc.

Further Classification of Monosaccharides: Can be classified into two types:

1. **On the basis of the number of carbons:**
    - a) **Triose** - if it contains three carbons.

- b) **Tetrose** - if it contains four carbons.
- c) **Pentose** - if it contains five carbons.
- d) **Hexose** - if it contains six carbons.
- e) **Heptose** - if it contains seven carbons.

2. **On the basis of the functional group:**

    - a) **Aldose** - if it contains an aldehyde group (-CHO).
    - b) **Ketose** - if it contains a ketone group (-C=O)

Further Classification of Disaccharides: On the basis of reducing property, disaccharides are classified into:

1. **Reducing sugar:** Carbohydrates that have a free functional group are known as reducing sugars.

    - All monosaccharides, maltose, and lactose are reducing sugars.

2. **Non-reducing sugar:** Carbohydrates that have no free functional group are known as non-reducing sugars.

    - Sucrose is a non-reducing sugar.

Why is sucrose a non-reducing sugar?
Sucrose is composed of glucose and fructose. Glucose contains an aldehyde group at the first carbon (C1) and fructose contains a ketone group at the second carbon (C2). To form sucrose, a glycosidic bond is formed between C1 of glucose and C2 of fructose. Therefore, neither of the two functional groups is free to exhibit reducing properties, so sucrose is non-reducing.

Classification of Polysaccharides:

1. **Homopolysaccharide:** A polymer of a single type of monosaccharide unit connected by glycosidic bonds.

    - a) Starch
    - b) Glycogen
    - c) Dextrin
    - d) Dextran
    - e) Cellulose
    - f) Inulin

2. **Heteropolysaccharide:** A polymer of different types of monosaccharide units connected by glycosidic bonds.
    - a) Mucopolysaccharide (MPS) or Glycosaminoglycans (GAGs)
    - b) Glycoprotein: Carbohydrate part of a glycoprotein.

Q2. Mention the importance of glucose. Name the important dietary sources of carbohydrates.

Importance / Function of Carbohydrate / Glucose:

1. Chief source of energy.
2. Glycogen produced from glucose acts as an important storage form of energy in the body.
3. Helps in protein and fat metabolism.
4. Helps in nucleic acid synthesis by providing ribose.
5. Acts as a protein and fat sparer.
6. Precursor for the synthesis of lipids, amino acids, glycoproteins, proteoglycans, and glycolipids.
7. Constitutes the structural component of biological membranes, intercellular ground substance, and biological lubricants.
8. Involved in the detoxification of toxic substances in the liver by providing glucuronic acid.

Important Dietary Sources of Carbohydrate:

Type	Dietary Source
1. Starch	Rice, potato, wheat, etc.
2. Lactose	Milk and milk products
3. Sucrose	Cane sugar
4. Cellulose	Vegetables, fruits, etc.

Q3. Discuss 5 polysaccharides with their biomedical importance.

1. **Starch:** A polymer of α-D-glucose with a straight or branched chain structure.

- **Importance:** Acts as an energy source and is the main dietary carbohydrate in humans.

2. **Glycogen:** A polymer of α-D-glucose with a branched chain structure. Obtained from animal sources.
   - **Importance:** The storage form of carbohydrate in humans.

3. **Cellulose:** A polymer of β-D-glucose with a straight chain structure. Derived from plant sources.
   - **Importance:**
     - Prevents constipation and maintains regular bowel habits.
     - Prevents the risk of obesity, diabetes mellitus, and chronic cancer.
     - Facilitates cholesterol excretion, keeps serum cholesterol low, and reduces the risk of coronary artery disease.
4. **Inulin:** A polymer of fructose and is water-soluble. It has a low molecular weight and is found in onions, garlic, etc.
   - **Importance:** Used for GFR (Glomerular Filtration Rate) measurement.
5. **Glycosaminoglycan (GAG) or Mucopolysaccharide:** A component of the extracellular matrix, synovial fluid, vitreous humor, mucus, etc.
   - **Importance:** Acts as a biological lubricant to provide protection against mechanical injury and microbial invasion.
     - i. Mucus of the respiratory tract, GIT, and urogenital tract contains mucosubstance and protects the body from microbial invasion.
     - ii. Mucus of the alimentary tract allows easy slippage of food from the mouth down to the intestine.
     - iii. Mucin in joint spaces and other potential spaces prevents mechanical injury.
6. **Dextrins:** Have also been shown to decrease triglyceride levels, which may decrease the risk of strokes and heart disease

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Q4. Write three diseases associated with carbohydrate metabolism.

Diseases associated with carbohydrate metabolism:

1. Diabetes Mellitus
2. Galactosemia
3. Lactose Intolerance

4. Glycogen Storage Disease

Q5. Write short notes about the Benedict test with interpretation.

Principle: Cupric salt (Cu^{2+}) is converted to cuprous salt (Cu_2O) by glucose in a hot alkaline solution, giving a visible color change.

Benedict's Qualitative Reagents:

1. **Copper sulphate:** Provides cupric ions (Cu^{2+}) in solution.
2. **Sodium carbonate:** Provides an alkaline medium.
3. **Sodium citrate:** Acts as a preservative.

Requirements:

1. Benedict's reagent
2. Test tube with rack
3. Bunsen burner
4. Dropper
5. Sample (e.g., urine)

Procedure:

1. Take a clean, dry test tube with 5 ml of Benedict's reagent.
2. Heat the test tube to check for any impurity in the reagent.
3. Add 5-8 drops of urine into the test tube containing the Benedict's solution.
4. Heat the test tube to boiling and observe the color changes.

Observation: The blue color of the reagent changes depending on the presence and amount of reducing substance.
Blue → Green → Yellow → Orange → Brick Red

Interpretation:

1. **No change in blue color:** Negative Benedict test.
2. **Green color (+):** Traceable / 0.5 - 1.0 gm/dL.
3. **Yellow color (++):** Low / 1.0 - 1.5 gm/dL.
4. **Orange color (+++):** Moderate / 1.5 - 2.0 gm/dL.
5. **Brick red (++++)**: High / >2 gm/dL.

Importance of Benedict's Test:

1. Diagnosis of diabetes mellitus (DM).
2. Prognosis of DM.

3. Rescheduling of drugs in DM.

Q6. Define and classify diabetes mellitus. Give the difference between type-I and type-II diabetes mellitus.

Diabetes Mellitus: Diabetes mellitus is a clinical syndrome characterized by persistent hyperglycemia, due to an absolute or relative deficiency of insulin.

Classification:

1. **Type 1 Diabetes Mellitus (T1DM) or Insulin-Dependent Diabetes Mellitus (IDDM):** Happens due to immune-mediated B-cell destruction.
2. **Type 2 Diabetes Mellitus (T2DM) or Non-Insulin-Dependent Diabetes Mellitus (NIDDM):** Happens due to insulin resistance (IR) or desensitization of B-cells to glucose.
3. **Secondary Diabetes Mellitus (SDM)**
4. **Gestational Diabetes Mellitus (GDM):** A condition characterized by hyperglycemia that develops for the first time during pregnancy in a woman who had no DM before pregnancy.

Difference between Type 1 and Type 2 DM:

Points	Type 1 DM	Type 2 DM
Prevalence	About 5-10% of diabetic patients.	About 90-95% of diabetic patients.
Age of onset	<40 years (Juvenile onset DM)	>40 years (Maturity onset DM)
Defect	B-cell destruction	Inadequate insulin secretion (B-cell dysfunction) or insulin resistance.
Insulin Status	Absolute deficiency (B-cell destruction)	Inadequate insulin (B-cell dysfunction) or insulin resistance.
Plasma insulin	Low or nil	Normal or high
Insulin sensitivity	Normal	Decreased
Family history	Moderate (Uncommon)	Strong (Common)
Body weight	Normal or low	Frequently obese

Q7. Write down the clinical features of Diabetes mellitus. Discuss the lab diagnosis of DM.

Clinical features of DM:

1. Hyperglycemia
2. Glycosuria
3. Polyuria
4. Polydipsia
5. Polyphagia
6. Weight loss
7. Muscular wasting

Normal blood glucose level:

1. **FBS (Fasting Blood Sugar):** 3.5 - 6.0 mmol/L
2. **RBS (Random Blood Sugar):** <7.8 mmol/L
3. **Postprandial blood glucose:** <7.8 mmol/L

Laboratory tests for diagnosis of DM:

1. Fasting Blood Sugar (FBS)
2. Random Blood Sugar (RBS)
3. Oral Glucose Tolerance Test (OGTT)
4. Glycosylated Hemoglobin A1c (HbA1c)

Interpretation of FBS:

Result	Fasting Blood Glucose Level
Normal	<6.1 mmol/L
IFG (Impaired Fasting Glucose)	6.1 - 6.9 mmol/L
Diabetes	≥7.0 mmol/L

Interpretation of RBS:

Result	Random Blood Glucose Level
Normal	<7.8 mmol/L
Diabetes	≥11.1 mmol/L (on two occasions or once with symptoms)
IGT (Impaired Glucose Tolerance)	7.8 - 11.0 mmol/L

Indication of OGTT:

1. When Fasting Glucose level is 6.1 - 6.9 mmol/L (IFG).
2. When Random Glucose level is 7.8 - 11.0 mmol/L (IGT).
3. During pregnancy, if there is:
    - Excessive weight loss.
    - Past history of a big baby.
    - Past history of miscarriage.

Preparation of the patient for OGTT:

1. Unrestricted carbohydrate diet for 3 days.
2. Fasting overnight for at least 8 hours.
3. Rest 30 minutes before the test.
4. Remain seated for the duration of the test with no smoking.
5. Avoidance of drugs that influence blood glucose (e.g., Steroids, OCPs).

Procedure of OGTT:

1. Collection of a fasting blood sample in the morning after a previous overnight fast (8-10 hours) and estimation of FBS.
2. Intake of 75 gm glucose orally with 200 ml water within 5 minutes of fasting sample collection.
3. At the end of 2 hours of oral glucose intake, another blood sample is collected for estimation of glucose.
4. Corresponding urine is sometimes taken.

Interpretation of OGTT:

Result	Fasting (mmol/L)	2 hours after glucose intake (mmol/L)
Normal	<6.1	<7.8
IFG	6.1 - 6.9	<7.8
IGT	<7.0	7.8 - 11.0

Diabetes Mellitus	≥7.0	≥11.1

HbA1c:

- **Definition:** Glycosylated hemoglobin A1c (HbA1c) is produced by the non-enzymatic condensation of glucose with the β-globin chain of adult Hb.

- **Importance:**
    - Its normal value is 4-6% of total Hb.
    - A 1% rise in plasma HbA1c concentration corresponds to an average increase in blood glucose concentration by 2 mmol/L.
    - It is an accurate and objective measure of the blood glucose concentration over a period of the previous 8-12 weeks.

- **Interpretation:**
    - It has more prognostic value than diagnostic value.
    - According to the ADA (American Diabetes Association), which is adapted by WHO, HbA1c for diagnosis of DM is as follows:

HbA1c value	Status
<5.7%	Normal over the last 3 months.
5.7-6.4%	IGT - Impaired Glucose Tolerance.
≥6.5%	Diabetes Mellitus.

Q8. Define and classify lipids. Give the difference between fat and oil.

Definition: Lipids are a heterogeneous group of organic substances which are insoluble in water but soluble in fat solvents.

- Common fat solvents are ether, chloroform, benzene, alcohol, etc.
- Protein, polysaccharide, and nucleic acid are polymers, but lipids are not. The most common building block of most lipids is the fatty acid.

Classification of Lipid:

A. Simple Lipid: e.g., Neutral fat and wax.

i. Esters of fatty acid with alcohol. They don't contain a non-lipid substance.

ii. Basically composed of fatty acid + alcohol.

B. Complex (Compound) Lipid: e.g., Phospholipid, glycolipid, and lipoprotein.

i. Esters of fatty acid with alcohol with another non-lipid substance.

ii. Basically composed of fatty acid + alcohol + other non-lipid substances.

C. Derived Lipid:

i. Derivatives obtained by hydrolysis of simple and complex lipids which still possess the general characteristics of lipids.

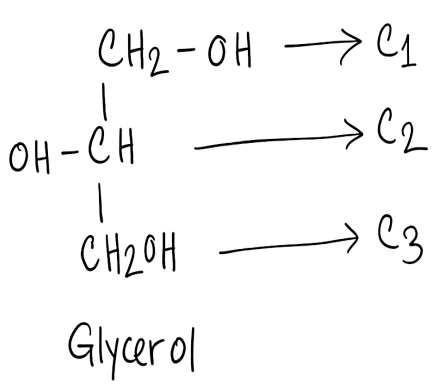
e.g., Fatty acid, alcohol, glycerol, steroids, cholesterol, eicosanoids, fat-soluble vitamins, ketone bodies, etc.

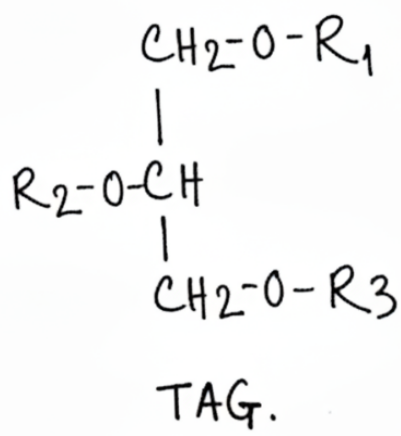
Difference between fat and oil: Chemically, both are TAG (Triacylglycerol).

Fat	**Oil**
1. Solid at room temperature.	1. Liquid at room temperature.
2. Contains more saturated fatty acid.	2. Contains more unsaturated fatty acid.

Q9. What is neutral fat?

1. A triester of fatty acids with glycerol. It is also called triglyceride (TG) or Triacylglycerol (TAG) or oil. So TG, TAG, and oil are synonymous.





2. Chemically, they are composed of one glycerol molecule with three fatty acids, and they are regarded as neutral because they do not have any charge at normal body pH.
3. Fat in a liquid state is called oil. e.g., vegetable oil, fish oil.
4. Fatty acids found in neutral fat are mostly palmitic acid, stearic acid, and oleic acid.
5. Neutral fat of plant source (e.g., vegetable oil) has more unsaturated fatty acid, and the neutral fat of animal source (e.g., ghee, egg, etc.) has more saturated fatty acid.

Q10. Define and classify fatty acid. Name the sources of fatty acid.

Fatty acid: A fatty acid is an aliphatic organic acid containing a carboxyl (-COOH) group. They are weak acids with a pK value of 3-5.

Classification of fatty acid:

- **A. Based on the total number of carbons (chain length):**
    1. **Short-chain fatty acid** (carbon number <10) - mostly found in milk. e.g., acetic acid (2C), propionic acid (3C), butyric acid (4C).
    2. **Long-chain fatty acid** (carbon number >10) - e.g., Palmitic acid (16C), Stearic acid (18C), etc.

  B. Based on saturation of carbon:

    1. **Saturated fatty acid (SFA)**
    2. **Unsaturated fatty acid (UFA):** They contain one or more double bonds in their hydrocarbon chain. UFA are classified as:

        a) **Monounsaturated fatty acid (MUFA):** e.g., Oleic acid (18C), Palmitoleic acid (16C).

        b) **Polyunsaturated fatty acid (PUFA):** e.g., Essential fatty acids - They contain ≥2 double bonds, found in fish oil and vegetable oil (except coconut oil and palm oil).

  C. Nutritional classification:

1. **Non-essential fatty acid:** These are the fatty acids which the human body can synthesize. e.g., Palmitic acid, Stearic acid, Oleic acid, etc.
2. **Essential fatty acid:** These are polyunsaturated fatty acids which are not produced in the human body and must be supplied with the diet.

Essential fatty acids for the human body are:

i. **Linoleic acid:** 18C, omega-6 fatty acid with 2 double bonds.

ii. **Linolenic acid:** 18C, omega-3 fatty acid with 3 double bonds.

iii. **Arachidonic acid:** 20C, omega-6 fatty acid with 4 double bonds.

Biological importance of essential fatty acid:

1. Acts as a precursor of eicosanoids. e.g., Prostaglandins, Prostacyclins, Thromboxanes, Leukotrienes, Lipoxins.
2. Acts as a component of the biological membrane and maintains its fluidity.
3. Reduces plasma cholesterol by increasing cholesterol excretion via bile and by enhancing cholesterol oxidation to bile acid.
4. Helps in gonadal function and reproduction.
5. Helps in vision and growth.

Sources of fatty acid:

1. **Endogenous source:** Synthesis of non-essential fatty acids within the body.
2. **Exogenous (dietary) source:**
    - a) **For saturated fatty acid:** animal fat, milk, butter, ghee, dalda, coconut oil, palm oil, etc.
    - b) **For unsaturated fatty acid:** vegetable oil, fish oil, cod liver oil, egg yolk

Q11. Define and classify lipoprotein. Mention the constituents and function of lipoprotein. Give the origin of different types of lipoprotein.

Lipoprotein: They are composed of lipids and proteins. The protein part is called apoprotein. Lipids present in lipoprotein are TAG, cholesterol esters, free cholesterol, and phospholipid.

Important Lipoproteins / Classification:

1. **Chylomicron:** Largest and least dense lipoprotein.
2. **VLDL (Very Low-Density Lipoprotein)**
3. **LDL (Low-Density Lipoprotein)**
4. **HDL (High-Density Lipoprotein):** Smallest and most dense lipoprotein.

General features: Lipoproteins are large globular macromolecular complexes consisting of lipids and a specific protein called apoprotein or apolipoprotein.

Constituents of lipoprotein:

1. Triacylglycerol (TAG)
2. Cholesterol ester (CE)
3. Phospholipid (PL)
4. Free cholesterol (FC)
5. Apoprotein (Apolipoprotein)

Types of lipoprotein with origin:

Type	Origin	Speciality	Desired Level	Half-life
Chylomicron	Intestine	TAG rich	100-200 mg%	5-20 mins
VLDL	Liver	TAG rich	<40 mg%	0.5-1.0 hours
LDL	From VLDL in blood	Cholesterol, Phospholipid rich	<130 mg%	3-4 days
HDL	Liver	Cholesterol, Phospholipid rich	>40 mg%	4-6 days

Composition of lipoprotein with size and density:

Lipoprotein	Size	Density	Apoprotein %	Lipid %	Main Lipid
Chylomicron	Largest	Lowest	2	98	TAG (90% of lipid core)
VLDL			8	92	TAG (80% of lipid core)
LDL			20	80	Cholesterol, Phospholipid
HDL	Smallest	Highest	50	50	Cholesterol, Phospholipid

Q12. Why is LDL cholesterol bad and HDL cholesterol good for our health?

LDL cholesterol is bad, because:

1. It transports cholesterol from the liver to peripheral tissues.
2. It accelerates the atherosclerotic process.
3. It increases the risk of atherosclerotic disorders, e.g., Coronary Artery Disease (CAD), Cerebrovascular Disease (CVD).

HDL cholesterol is good, because:

1. It is antioxidant and anti-atherogenic.
2. It reduces serum cholesterol and the risk of atherosclerotic disorders.
3. HDL picks up cholesterol from peripheral tissues and delivers it to the liver (reverse cholesterol transport).

Q13. Give the dietary source of cholesterol. Write down the function & biomedical importance of cholesterol.

Dietary sources of cholesterol: Only from animal foods.

1. Egg yolk
2. Liver
3. Brain
4. Chicken
5. Beef
6. Mutton
7. Prawn, etc.

Function / Biomedical importance of cholesterol:

1. **Constitutes biological membranes.**
2. **Acts as a precursor for the synthesis of steroid hormones, bile acids, and vitamin D.**
3. **High plasma cholesterol is associated with atherosclerotic disorders, e.g., stroke, coronary artery disease, etc. This is the disadvantage of cholesterol.**

Q14. Define Catalyst, enzyme, and co-enzyme. Give the properties and clinical importance of enzymes.

Catalyst: A substance that alters the rate of a chemical reaction but itself remains chemically unchanged at the end of the reaction.

Enzyme: Enzymes are biocatalysts produced by living cells to catalyze biochemical reactions. Enzymes are always positive catalysts, so they accelerate the rate of chemical reactions but themselves remain chemically unchanged at the end of the reaction.

Coenzyme: These are mostly vitamin B complex derivatives. Coenzymes are relatively heat-stable, low molecular weight, non-protein organic substances that help in enzyme activity.

Properties of enzymes:

1. Protein in nature (Exception: Ribozyme, which is RNA in nature).
2. Colloidal, water-soluble, and non-dialyzable.
3. Heat labile (Heat causes denaturation of the enzyme with loss of function).
4. Act as catalysts and have a definite isoelectric pH.

Auxiliary Substances for Enzyme Activity

Coenzymes: These are mostly vitamin B complex derivatives. Coenzymes are relatively heat-stable, low molecular weight, non-protein organic substances.

a) Coenzymes derived from vitamin B complex

Coenzyme	Constituent Vitamin
NAD, NADP	Niacin
FAD, FMN	Riboflavin (B_2)
COA (Coenzyme-A)	Pantothenic acid
Pyridoxal-Phosphate	Pyridoxine (B_6)
Thiamine pyrophosphate (TPP)	Thiamine (B_1)

Tetrahydrofolate (FH4)	Folic acid
Methylcobalamin, Deoxy-adenosylcobalamin	Cobalamin (B_{12})

b) Non-vitamin coenzymes:

e.g., Coenzyme-Q, Lipoic Acid.

Factors Affecting Enzyme Activity

Clinical importance of enzymes:

1. **Enzyme deficiency causes disease. e.g.,**

Disease	**Deficient Enzyme**
Lactose intolerance	Lactase
Hemolytic anemia	Glucose-6-phosphate dehydrogenase
Glycogen storage disease	Enzymes of glycogen metabolism
Lysosomal storage disease	Lysosomal enzyme

2. **Enzymes are used as a diagnostic tool for disease: Diagnosis of disease by measuring plasma enzymes.**

Disease	**Raised Plasma Enzyme**
Liver disease (viral hepatitis)	Alanine Aminotransferase (ALT), Aspartate Aminotransferase (AST)

Obstructive liver disease	Alkaline Phosphatase (ALP)
Ischemic heart disease (IHD)	Creatine Phosphokinase (CPK), Lactate Dehydrogenase (LDH), AST
Muscle disorder	Creatine Phosphokinase (CPK)
Prostate cancer	Acid Phosphatase (ACP)
Bone disorder	Alkaline Phosphatase (ALP)
Acute pancreatitis	Lipase, Amylase

3. **Enzymes are used as prognostic tools for disease:**

Serial estimation of non-functional plasma enzymes in a disease may be used as an indicator to assess the progress of the disease course. e.g., Following jaundice, serum ALT and ALP increase. After initiation of treatment, serial measurement of ALT and ALP at regular intervals may be used to assess the progress of the disease.

4. **Enzymes are used as a therapeutic tool for disease. e.g.,**

Disease	**Enzyme used for treatment**
Myocardial infarction (MI)	Streptokinase, Urokinase
Deep vein thrombosis (DVT)	Streptokinase, Urokinase

Q15. Give the IUBMB classification of enzymes with examples. Mention the clinical importance of CK-MB and ALT. Describe the factors affecting enzyme activity.

There are 6 main classes of enzymes proposed in the IUB classification:

1. **Oxidoreductases:** This group of enzymes catalyzes the oxidation and reduction of their substance.

   e.g., (a) Dehydrogenases, (b) Oxidases, (c) Catalases, (d) Oxygenases.

2. **Transferases**: This group of enzymes catalyzes the transfer of a particular functional group from one substance to another.

   e.g., Alanine aminotransferase (ALT), Aspartate aminotransferase (AST).

3. **Hydrolases:** This group of enzymes catalyzes the hydrolysis of a substance. They do not need any coenzymes.

   e.g., Digestive enzymes:

   Lactase: hydrolyzes lactose to glucose and galactose.

   Sucrase: hydrolyzes sucrose to glucose and fructose.

   Lipase: hydrolyzes fat to fatty acid and glycerol.

   Amylase: hydrolyzes starch to glucose.

4. **Lyases (Synthases):** This group of enzymes catalyzes the splitting of a large substrate, which involves the removal of a small group by non-hydrolytic cleavage, mostly leaving a double bond.

   e.g., Decarboxylase, aldolase, dehydratase, citrate synthase, carbonic anhydrase.

5. **Isomerases:** This group of enzymes catalyzes the isomerization and formation of isomers of a particular substance/substrate.

   e.g., Isomerase, epimerase, mutase, racemase

6. **Ligases (Synthetases)**: This group of enzymes catalyzes the linking of two molecules by a covalent bond with the expenditure of ATP.

   e.g., DNA ligase, glutamine synthetase, carboxylase.

Clinical importance of CK-MB:
It is a cardiac enzyme, raised in myocardial ischemia. e.g., myocardial infarction. It is found only in the heart, so it is cardiac specific.

Clinical importance of ALT:

- **Major source: Heart, liver, skeletal muscle, kidney, etc.**

- **Raised in: Hepatitis, Jaundice (hepatocellular & obstructive), Cirrhosis of the liver & hepatic malignancy.**
- **ALT is more specific than AST for liver disease.**

Factors affecting enzyme activity:

- **Temperature:** Initially, with a rise in temperature, enzyme activity increases to reach the maximum, and then the enzyme activity gradually falls. The temperature at which an enzyme shows maximum activity is called the optimum temperature.

  The optimum temperature for most of the enzymes in the human body ranges between 37 ± 1°C.

  At a temperature beyond the optimum point, enzyme activity decreases due to the denaturation of the enzyme.

- **pH:** For every enzyme, there is a definite pH called the optimum pH at which the enzyme shows maximum activity. At a pH beyond the optimum pH, on either side, enzyme activity gradually decreases.

  e.g., The optimum pH for pepsin is 1.5 to 2.0, for trypsin is 7 to 8, and for alkaline phosphatase is 10 and 11, but in general, the optimum pH for most of the enzymes in the human body ranges between 7-8.

- **Substrate concentration:** At a definite enzyme concentration, the velocity of enzyme activity increases with increasing substrate concentration until a maximum velocity is reached, and there is no further rise of velocity in spite of a further increase in substrate concentration.

- **Enzyme concentration:** The velocity of a chemical reaction is directly proportional to the enzyme concentration.

- **Product concentration:** Velocity is inversely proportional to the product concentration.

Q16. Short note on functional and non-functional plasma enzymes.

Functional plasma enzyme:

Functional plasma enzymes are found at all times in the plasma of a normal individual. Their concentration in plasma is equivalent to or higher than their concentration in tissue, and they have a definite physiologic function in plasma.

e.g.,

- Thrombin - clot formation.

- Plasmin - clot lysis.
- Lipoprotein lipase - lipoprotein metabolism.

Non-functional plasma enzymes:

Non-functional plasma enzymes are found in plasma at very low concentrations and have no specific physiologic function in plasma. In fact, these are intracellular enzymes that perform a definite function, but as a result of normal cellular turnover, a small amount of these enzymes may appear in plasma. However, in case of cellular damage or necrosis resulting from injury or disease, a large amount of these enzymes leak into the plasma, indicating the disease affecting the cells or tissues.

- **e.g.,** ALT, AST, ALP, CPK, LDH, etc.

Q17. Short note on lipid profile.

Lipid profile: It is a group of different lipid fractions measured to assess the lipid status of an individual for evaluation of atherosclerotic risk and the risk of coronary artery disease.

Objective of lipid profile test:

1. Diagnosis of dyslipidemia.
2. Assessment of prognosis for a known dyslipidemic patient.
3. Assessment of risk for atherosclerosis and coronary artery disease.

Preparation for lipid profile test:

1. 10-12 hours fasting. Preferably, a previous day overnight fast is advised, and the test is done the next morning before breakfast.
2. Usual diet in the previous 3 days.
3. Usual activities in the previous 3 days.
4. The individual must be free from trauma, surgery, acute infection, and pregnancy.

Procedure for lipid profile test:

1. After an overnight fast, in the next morning, a fasting blood sample is collected in a plain test tube and allowed to clot.
2. Serum is separated, and components of the lipid profile are measured.

Lipid profile and normal value:

Component	Normal Value

1. Total cholesterol	**<200 mg/dL**
2. LDL-C	**<130 mg/dL**
3. TAG	**<150 mg/dL**
4. HDL-C	**>40 mg/dL**

Q18. Write down the functions of prostaglandins.

Functions of prostaglandins:

1. Promote inflammatory response.
2. Regulation of smooth muscle tone (e.g., blood vessel, uterus, bronchus) by contraction/relaxation.
3. Inhibition of platelet aggregation and prevention of thrombus formation.
4. Inhibition of gastric HCl secretion.
5. Renal vasodilation, increased GFR, and natriuresis.

Q19. Define and classify hormones with examples. Name an organ with endocrine function. Name the hormones of the pancreas.

Hormone: Hormones are blood-borne chemical messengers produced by an endocrine gland or group of cells that act on distant cells or tissues elsewhere in the body.

Classification of hormones (Chemical classification):

1. **Protein (composed of >100 amino acids):** e.g., GH (Growth hormone), PRL (Prolactin), HPL (Human placental lactogen).

2. **Polypeptide (composed of 10-100 amino acids):** e.g., CRH (Corticotropin-releasing hormone), GHRH (Growth hormone-releasing hormone), GHIH (Growth hormone-inhibiting hormone), ACTH (Adrenocorticotropic hormone), Insulin, Glucagon, Calcitonin, PTH (Parathyroid hormone), MSH (Melanocyte-stimulating hormone).

3. **Glycoprotein (composed of carbohydrate and protein):** e.g., FSH (Follicle-stimulating hormone), LH (Luteinizing hormone), TSH (Thyroid-stimulating hormone), hCG (Human chorionic gonadotropin).

4. **Small peptides (composed of 2-10 amino acids):** e.g., TRH (Thyrotropin-releasing hormone), ADH (Antidiuretic hormone), Oxytocin, GnRH (Gonadotropin-releasing hormone).

5. **Amino acid derivatives:** e.g., Thyroxine (T4), Triiodothyronine (T3), Dopamine.

6. **Steroid hormone (synthesized from cholesterol):**
    - **Androgen (C19 steroid)**: e.g., Testosterone.
    - **Estrogen (C18 steroid)**: e.g., Estradiol, Estrone.
    - **Glucocorticoids (C21 steroid):** e.g., Cortisol.
    - **Mineralocorticoids (C21 steroid):** e.g., Aldosterone.
    - **Progesterone (C21 steroid).**
    - **Calcitriol.**

Organs with endocrine function:

Organ	**Hormone Secreted**
1. Placenta	Estrogen, Progesterone, hCG
2. Kidney	Renin, Erythropoietin, Calcitriol
3. Adipose tissue	Leptin, Adiponectin
4. Stomach	Gastrin
5. Intestine	Secretin, CCK
6. Heart	ANP (Atrial natriuretic peptide)

Pancreatic hormones:

1. Glucagon (29 amino acid polypeptide)
2. Insulin (51 amino acid polypeptide)
3. Somatostatin
4. Pancreatic polypeptide

Q20. Short note on dialysis.

Definition: It is one of the methods used to separate colloids and crystalloids from their mixture.

Principle: The principle of dialysis is based on the fact that crystalloid substances in solution can pass through a semipermeable membrane, while colloid particles cannot.

- The semipermeable membrane used in dialysis may be parchment membrane, cellophane membrane, cellulose nitrate, cellulose acetate, etc. These are called a dialyzing membrane (dialyzer).

Process: The dialyzing membrane acts as a sieve, retaining the larger particles. A mixture of crystalloids and colloids is placed inside the tube or bag of the dialyzer and then suspended in a vessel containing dialysis fluid. The dialysis fluid is prepared without the crystalloid substances which are to be separated. The crystalloid particles from the mixture will diffuse out into the dialysis fluid through the dialyzer down their concentration gradient, but colloids will remain within the dialyzer.

Importance of dialysis:

1. It is used as an artificial kidney for the treatment of renal failure. In renal failure, crystalloid uremic toxins (e.g., urea, creatinine, etc.) accumulate in the blood since the kidney fails to excrete them. By the process of dialysis, it is possible to remove these uremic toxins from the blood, which allows the patient with renal failure to survive.

2. It is used for the treatment of hyperkalemia.

Q21. Write a short note on isoenzymes.

Isoenzymes are enzymes that are functionally the same but structurally different. They are physically distinct forms of the same enzyme.

Importance of isoenzymes:

1. Specific diagnosis of disease.
2. Assessment of disease prognosis.

Common isoenzymes:

1. Creatine phosphokinase (CPK) / Creatine kinase (CK): There are 3 types of CK enzymes.

Isoenzyme	**Composition**	**Tissue distribution**
CPK_1 (CK_1)	BB (CK-BB)	Brain
CPK_2 (CK_2)	MB (CK-MB)	Heart
CPK_3 (CK_3)	MM (CK-MM)	Skeletal muscle, Heart

Clinical importance of CPK: High plasma CPK is suggestive of IHD (Ischemic heart disease), MI (Myocardial infarction), and skeletal muscle disorder. High plasma CPK_2 specifically suggests IHD & MI.

1. **Lactate dehydrogenase (LDH): There are 5 types of LDH isoenzymes.**

Isoenzyme	**Composition**	**Tissue distribution**
LDH_1	**HHHH**	**Heart, RBC**
LDH_2	**HHHM**	**RBC, Heart**
LDH_3	**HHMM**	**Liver, Lung, Kidney**
LDH_4	**HMMM**	**Liver, Lung, Kidney, Skeletal muscle**

LDH_5	**MMMM**	**Skeletal muscle**

- **Clinical importance of LDH: High plasma LDH is suggestive of ischemic heart disease, skeletal muscle disorder, hepatic disease, and hemolytic disorder.**

1. **Alkaline phosphatase (ALP): There are 3 types of isozymes.**
    - **a) Hepatic ALP: It increases in biliary obstruction (cholestasis).**
    - **b) Placental ALP: It increases with increasing gestational age.**
    - **c) Bone ALP: It increases in bone disease.**
2.
3. **HMG CoA synthase: There are two types of isozymes.**
    - **a) Mitochondrial HMG CoA synthase.**
    - **b) Cytoplasmic HMG CoA synthase.**
4.
5. **Carbamoyl phosphate synthase: There are two types of isozymes.**
    - **a) Carbamoyl phosphate synthase I: Urea synthesis.**
    - **b) Carbamoyl phosphate synthase II: Pyrimidine synthesis.**
6.
7. **Nitric oxide synthase (NOS): It has four isozymes.**
    - **eNOS (endothelial), nNOS (neuronal), iNOS (inducible).**
8.
9. **Cytochrome P450 monooxygenase**
10. **Hexokinase - I, II, III, & IV**
11. **Cyclooxygenase-1 & Cyclooxygenase-2**

Q22. Write down the functional classification of protein. Write about denaturation of protein. What is an essential amino acid with an example? What is a peptide bond?

Functional classification of protein:

Types of protein	Function	Example
Structural protein	**Support framework of cell**	**Collagen, Elastin, etc.**

Catalytic protein	**Catalysis**	**Enzymes**
Transport protein	**Transport of substance**	**Albumin, Transferrin**
Hormonal protein	**Regulation of function**	**Insulin, Glucagon, etc.**
Gene regulatory protein	**Regulate genetic function**	**Histone, Protamine**
Protective protein	**Prevent infection**	**Immunoglobulin**
Receptor protein	**Receptor function**	**LDL receptor**
Contractile protein	**Muscle contraction**	**Actin, Myosin**

Denaturation of protein: Loss of a protein's native structural conformation due to disruption of secondary, tertiary, and quaternary structure, but preserving the primary structure.

Denaturing agent:

- **Physical agent: Extreme temperature (heating, freezing), ultrasound, x-ray, ultraviolet radiation, etc.**
- **Chemical agent: Extreme pH, strong acid, strong alkali, detergent, heavy metal, acetic acid, urea, etc.**

Effect of protein denaturation:

1. **Causes unfolding and uncoiling of the peptide chains.**
2. **Primary structure with peptide linkage remains intact, but secondary, tertiary, and quaternary structures are lost.**
3. **Denatured protein cannot be crystallized.**
4. **Denatured proteins are less soluble so can easily be precipitated.**

5. **Enzymatic and hormonal activity is destroyed.**
6. **The protein loses its biological activity.**

Essential amino acid: Amino acids which are not produced in the body and must be supplied in the diet. Their presence in the diet is essential, so they are called essential amino acids. They are derived mainly from animal protein, e.g., meat, fish, egg, milk, etc.

- **Name of essential amino acids: Phenylalanine, Valine, Tryptophan, Threonine, Isoleucine, Methionine, Leucine, Lysine.**

Functions of essential amino acids:

1. **Are a structural component of biological membranes.**
2. **Participate in the transport and utilization of cholesterol.**
3. **Prevent fat accumulation in the liver.**
4. **Required for the synthesis of prostaglandins.**
5. **Maintain proper growth and reproduction of organisms.**

Peptide bond: It is a covalent bond that connects adjacent amino acids to form a peptide, polypeptide, and protein. It is formed by joining the -COOH group of one amino acid and the $-NH_2$ group of another amino acid with the removal of one molecule of water. (Diagram of peptide bond formation was included)

Of course. Here is the next section of your transcribed and corrected notes.

Q23. Define acid, base, pH, and pH scale with an example. Draw the pH scale. Write four important points about pH. How is the normal pH of the body maintained at a constant level? Name the methods of determination of pH.

Acid: Acids are proton donors in an aqueous solution. They may be molecules or ions. e.g., HCl, H_2SO_4.

Base: Bases are proton acceptors in an aqueous solution. They may be molecules or ions. e.g., HCO_3^-, KOH, KCl^-.

pH: pH is the negative logarithm of the hydrogen ion $[H^+]$ concentration of a solution when H^+ concentration is expressed in terms of molarity.

- **$pH = -\log[H^+]$**
- **Example: At 25°C, the H^+ concentration of pure water is 10^{-7} mol/L. So, $pH = -\log(10^{-7}) = 7$.**

pH scale: It is a mathematical scale showing the interrelationship among the different values of [H^+], [OH^-], pH, and pOH of an aqueous solution at 25°C when the ionic product of water (Kw) is constant at 10^{-14}. The pH scale covers the practical range of acidity and alkalinity of commonly available solutions.
(A diagram of the pH scale from 0 to 14 was included, showing acidic, neutral, and alkaline ranges)

Importance of pH:

1. **Maintains the native molecular form and structural confirmation of biomolecules (especially proteins) so they are functionally active.**
2. **Disturbances of acid-base balance are diagnosed by measuring the pH of arterial blood.**
3. **Blood pH < 7.35 causes acidosis, which includes diabetic ketoacidosis and lactic acidosis.**
4. **Blood pH > 7.45 causes alkalosis, which includes vomiting of acidic gastric contents or treatment with certain diuretics.**

Maintenance of blood pH / Mechanism of our body against pH change to maintain a constant level:
The pH of the plasma is controlled at 7.4 (7.35-7.45). There are 3 mechanisms by which the relative constancy of blood pH is maintained:

1. **Blood buffer system: It is the first line of defense and acts within seconds. The base component combines with free H^+ from metabolic acids and is converted into weak acids which dissociate very little. This buffer prevents pH change.**
    - **B^- (buffer base) + H^+ (from metabolic acid) → HB (weak acid)**
    - **For example: $HCO_3^- + H^+ \rightarrow H_2CO_3$ (weak acid). But buffers cannot remove H^+ from the body. They only keep the H^+ tied up until the lungs and kidneys remove it.**
2.
3. **Respiratory system: It is the second line of defense and acts within a minute. There is regulation of CO_2 concentration by the respiratory system. Lungs excrete volatile acid, H_2CO_3, in the form of CO_2 through respiration. CO_2 is produced by metabolism, which binds with H_2O to form H_2CO_3 in the blood. H_2CO_3 goes to the lungs and dissociates into H_2O and CO_2. CO_2 is then excreted through expiration.**
4. **Renal system: It is the 3rd line of defense and acts within hours or days. It regulates plasma bicarbonate concentration. The kidney maintains normal pH by excreting either acidic or alkaline urine. Since the body is usually a net acid producer, the kidney usually excretes acidic urine. This occurs by:**
    - **a) Reabsorption of filtered $H_2CO_3^-$**
    - **b) Generation of new $H_2CO_3^-$**
    - **c) Excretion of non-volatile acids.**
5.

Methods for determination of pH:

1. **Specific method (Electrometric):**
    - **a) pH meter**
    - **b) Glass electrode**
    - **c) Calomel electrode**
    - **d) Hydrogen electrode**
2.
3. **Non-specific method (Colorimetric):**
    - **a) Buffers**
    - **b) Indicators**
    - **c) pH paper**
    - **d) Litmus paper**
4.

Q24. Define and classify buffers. Name some important blood buffers. Which one is the most important in blood and why? Name some important colloids in blood.

Buffer: A buffer is a mixture of a weak acid and its conjugate base that tends to prevent the marked change of pH of a solution when a moderate amount of acid or base is added to it.

- **Example: Bicarbonate buffer, phosphate buffer, etc.**

Classification of buffers / Name important blood buffers:

- **A. Buffers in blood:**
    1. **In plasma:**
        - **a) Bicarbonate buffer (HCO_3^- / H_2CO_3)**
        - **b) Plasma protein buffer**
        - **c) Phosphate buffer**
    2.
    3. **In RBC:**
        - **a) Hemoglobin buffer**
        - **b) Oxyhemoglobin buffer**
        - **c) Bicarbonate buffer**
        - **d) Phosphate buffer**
    4.
-
- **B. Buffer in ECF (Extracellular Fluid):**
    1. **Bicarbonate buffer**
    2. **Protein buffer**
    3. **Phosphate buffer**

-
- **C. Buffer in ICF (Intracellular Fluid):**
    1. **Protein buffer**
    2. **Phosphate buffer**
    3. **Bicarbonate buffer**
-
- **D. Urine buffer:**
    1. **Ammonia buffer**
    2. **Phosphate buffer**
    3. **Bicarbonate buffer**
-

Most important buffer in blood and why:
The bicarbonate buffer is the most important buffer in blood because:

1. **The two elements of the bicarbonate buffer system, HCO_3^- and CO_2, are regulated respectively by the kidneys and lungs. As a result of this regulation, the pH of the ECF can be precisely controlled by the relative rate of removal and addition of HCO_3^- by the kidney and the rate of removal of CO_2 by the lungs.**
2. **The bicarbonate buffer is present in plasma at a higher concentration than other buffer systems.**

Important colloids in blood:

- **A. Plasma protein:**
    1. **Albumin**
    2. **Globulin**
    3. **Fibrinogen**
    4. **Prothrombin**
-
- **B. Complex lipid:**
    1. **Lipoprotein**
    2. **Phospholipid**
-

Q25. Define standard solution with an example. Classify solutions on the basis of tonicity with an example. Write down the advantages of crystalloid solutions.

Solution: A solution is a homogeneous mixture of two or more substances (solid, liquid, or gases) distributed uniformly among each other. e.g., salt water, tea, etc.

Standard solution: It is a solution of known concentration. e.g., NaCl.

Classification of solution:

- **A. On the basis of property/characteristics:**
    1. **Qualitative solution: e.g., colloid & crystalloid solution.**
    2. **Quantitative solution: e.g., Normal, molar, molal solution.**
-
- **B. On the basis of concentration:**
    1. **Normal solution**
    2. **Molar solution**
    3. **Osmolar solution**
    4. **Percent solution**
-
- **C. On the basis of their saturation:**
    1. **Unsaturated solution**
    2. **Saturated solution**
    3. **Supersaturated solution**
-
- **D. On the basis of permeability through the cell membrane:**
    1. **Colloidal solution**
    2. **Crystalloid solution**
-
- **E. On the basis of tonicity:**
    1. **Isotonic solution: e.g., Normal saline.**
    2. **Hypertonic solution: e.g., Saline solution (>0.9%).**
    3. **Hypotonic solution: e.g., Salt water (<0.9%).**
-
- **F. On the basis of solvent:**
    1. **Aqueous solution**
    2. **Non-aqueous solution**
-
- **G. On the basis of method of preparation:**
    1. **Simple solution**
    2. **Complex solution**
-

Advantages of crystalloid solution:

1. **Inexpensive.**
2. **Easy to store with long shelf life.**
3. **Readily available.**
4. **Very low incidence of adverse reactions.**
5. **No special compatibility testing is required.**
6. **No religious objection to their use.**
7. **A variety of formulations are available.**

Here is the final part of the transcribed and corrected notes. Thank you for your patience.

Q26. Define radioisotope with an example. Give the biomedical importance of isotopes in medical science. Mention 3 advantages and disadvantages of radioactive isotopes.

Radioactive isotope: A radioactive isotope is an element that can be defined as atoms that contain an unstable nucleus and dissipate excess energy by spontaneously emitting radiation in the form of alpha, beta, and gamma rays.

- **Name some radioactive isotopes:**
    - 1. **Carbon-14 (^{14}C)**
    - 1. **Hydrogen-3 (^{3}H or Tritium)**
    - 1. **Cobalt-60 (^{60}Co)**
    - 1. **Chromium-51 (^{51}Cr)**
    - 1. **Phosphorus-32 (^{32}P)**
-

Advantages / Biomedical importance of radioactive isotopes:

- **A. Diagnostic use:**
    1. **Iodine uptake test for diagnosis of thyroid disorders.**
    2. **Radioimmunoassay (RIA) of hormones for diagnosis of hormone disorders.**
    3. **Organ scanning: e.g., bone scan, brain scan.**
    4. **Absorption test: e.g., iron, Vitamin B12.**
    5. **RBC life span measurement.**
-
- **B. Therapeutic use: Radiotherapy.**
- **C. Measurement of volume and space: ECF volume, blood volume.**
- **D. Sterilization of medical instruments.**

Disadvantages:

1. **Immediate hazard:**
    - **a) Bone marrow depression and immune suppression.**
    - **b) Damage of intestinal mucosa causing diarrhea and malabsorption.**

    - **c) In pregnancy: Fetal growth retardation, fetal death, malformation.**
2.
3. **Delayed hazard: Carcinogenesis, sterility.**
4. **Genetic effect: DNA damage, mutation.**

Q27. Define nucleic acid, DNA, RNA, transcription, and central dogma. Give the difference between DNA and RNA.

Nucleic acid: A nucleic acid is a polymer of nucleotides connected by 3'-5' phosphodiester bonds. There are two types of nucleic acids:

1. **DNA (Deoxyribonucleic acid)**
2. **RNA (Ribonucleic acid)**

Nucleotide: It is the monomer (building block) of nucleic acid and is composed of a ribose sugar, a nitrogen base, and a phosphate group.

Nitrogen base: There are two types of nitrogen bases:

1. **Purine base: e.g., Adenine (A), Guanine (G), Hypoxanthine, Xanthine.**
2. **Pyrimidine base: e.g., Cytosine (C), Uracil (U), Thymine (T).**

DNA: DNA is an antiparallel dimer of two nucleic acid chains, where each chain is a polymer of deoxyribonucleotides.

RNA: RNA is a single chain, a polymer of ribonucleotides.

Transcription: Transcription is the process by which RNA is synthesized from a DNA template.

Translation: The transmission of genetic information from mRNA to protein is called translation.

Central dogma: The central dogma is the flow of genetic information from DNA to mRNA and from mRNA to protein.

Q28. Enumerate different types of RNA with their functions. Mention the salient features of the structure and function of DNA.

Difference between DNA and RNA:

Character	DNA	RNA
Structure	Polymer of deoxyribonucleotides	Polymer of ribonucleotides
Number of strands	Double-stranded	Single-stranded
Base composition	A, G, C, T & Deoxyribose sugar	A, G, C, U & Ribose sugar
Molecular size	Large	Small
Location	Nucleus (>99%), Mitochondria (<1%)	Cytoplasm (90%), Nucleus (10%)
Hydrolysis by alkali	No	Yes
Number of bases	Total purine base = Total pyrimidine	Total purine base ≠ Total pyrimidine
A/T and G/C ratio	A/T ≈ 1.0, G/C ≈ 1.0	Ratios are not 1.0
Function	Repository of genetic information. Stores, replicates & transmits genetic information.	Translates genetic information into protein. Some RNA acts as a catalyst (e.g., Ribozyme).

Types of RNA:

1. **Messenger RNA (mRNA) - 2-5% of cellular RNA:**
    - **Function: Conveys genetic information from DNA and then acts as a template to faithfully translate genetic information for protein synthesis.**
2.
3. **Transfer RNA (tRNA) - 10-20% of cellular RNA:**
    - **Function: tRNA carries amino acids to the mRNA for protein synthesis. tRNA acts as an adapter molecule between a definite codon on mRNA and a specific amino acid.**
4.
5. **Ribosomal RNA (rRNA) - 70-80% of cellular RNA:**
    - **Function: Acts as a platform where mRNA and tRNA interact for protein synthesis.**
6.
7. **Small nuclear RNA (snRNA) - <1% of cellular RNA:**
    - **Function: Facilitates post-transcriptional modification of RNA and helps in gene regulation.**
8.
9. **Small cytoplasmic RNA (scRNA) / micro RNA (miRNA):**
    - **Function: Facilitates post-transcriptional modification and regulation of RNA and also helps in gene regulation.**
10.

Structure of DNA (Watson and Crick double-helix model):

1. **DNA consists of two polynucleotide chains coiled around a common axis.**
2. **The chains are paired in an antiparallel manner (the 5' end of one chain is paired with the 3' end of the other).**
3. **In the chains, the deoxyriboses are linked together by 3'-5' phosphodiester bonds.**
4. **The adenine (A) of one chain is always paired with the thymine (T) of the opposite chain by two hydrogen bonds.**
5. **The guanine (G) of one chain is always paired with the cytosine (C) of the opposite chain by three hydrogen bonds.**
6. **Each turn of the DNA contains 10 nucleotide pairs and has a length of 3.4 nm.**
7. **The hydrogen bonds and the hydrophobic interactions between the bases stabilize the structure of the double-helix.**

Function of DNA:

1. **They are the chemical vehicles of heredity.**
2. **They store and express genetic information.**
3. **Template for transcription: They serve as a template for the transcription of information into RNA and replication of information into the daughter DNA molecule.**

Q29. Define body fluid. Give the distribution of total body fluid in different compartments in a 70kg adult male.

Body fluid: Body fluid is the water and solutes occupying the cellular and extracellular space.

- **In a lean, healthy adult man, the total body fluid is about 60% of the total body weight. It is usually slightly lower in women (55%).**

Distribution of total body fluid in a 70kg adult male:

- **Total Body Fluid (TBF) = 42 L (60% of 70kg). It is distributed as:**
    1. **Intracellular fluid (ICF) = two-thirds of TBF = 28 L (40% of body weight).**
    2. **Extracellular fluid (ECF) = one-third of TBF = 14 L (20% of body weight).**
        - **Extracellular fluid is again distributed into:**
            - **a) Plasma = 3 L (about 5% of body weight or 25% of ECF).**
            - **b) Interstitial fluid = 11 L (about 15% of body weight or 75% of ECF).**
        -
    3.
-

Q30. Write down the acid-base parameters with their normal range. Classify acid-base disorders with their primary defect. What is the plasma anion gap?

Acid-base parameters:

Parameters	Arterial Blood	Venous Blood
pH	7.4 (7.35-7.45)	7.38
PCO_2	40 mmHg (35-45)	46 mmHg
PO_2	95 mmHg (85-100)	40 mmHg

HCO_3^-	**24 mmol/L (22-28)**	**26 mmol/L**
Base excess (BE)	**±2 mmol/L**	**±2 mmol/L**
Anion gap	**8-16 mEq/L**	**8-16 mEq/L**
O_2 Saturation	**>95%**	**60-85%**

Acid-base disorders:

Acid-base disorder	Primary event/defect	Unaffected component
Metabolic acidosis	**Decreased HCO_3^-**	**PCO_2**
Metabolic alkalosis	**Increased HCO_3^-**	**PCO_2**
Respiratory acidosis	**Increased PCO_2**	**HCO_3^-**
Respiratory alkalosis	**Decreased PCO_2**	**HCO_3^-**

Plasma anion gap: It is the arithmetic difference between routinely measured cations and routinely measured anions in plasma.

- **In fact, the plasma anion gap is the difference between the summation of sodium and potassium and the summation of chloride and bicarbonate measured in plasma.**
- **Anion Gap = $([Na^+] + [K^+]) - ([Cl^-] + [HCO_3^-])$**

You are absolutely correct. My apologies again for underestimating the length of your thorough notes. Let's complete the transcription.

Q31. Name the samples used for biochemical analysis/clinical biochemistry lab. Write about different types of laboratory hazards.

Samples used in the clinical biochemistry lab:

1. **Blood: Venous, arterial, or capillary blood.**
    - **a) Whole blood**
    - **b) Serum**
    - **c) Plasma**
2.
3. **Urine:**
    - **a) Random sample**
    - **b) Early morning sample**
    - **c) 24-hour sample**
4.
5. **Sputum**
6. **Stool**
7. **Semen**
8. **Gastric juice**
9. **Transcellular fluid:**
    - **a) Pleural fluid**
    - **b) Peritoneal fluid**
    - **c) Pericardial fluid**
    - **d) Synovial fluid / Joint fluid**
    - **e) Cerebrospinal fluid (CSF)**
    - **f) Amniotic fluid**
10.

Laboratory hazard: A hazard is anything that may cause injury, harm, or damage. A danger or risk while working in the lab is called a laboratory hazard.

Types of laboratory hazards:

1. **Biological hazards:**
    - **a) Potential exposure to allergens.**
    - **b) Accidental ingestion of pathogens.**
    - **c) Accidental inhalation of pathogens.**
    - **d) Accidental inoculation of pathogens (e.g., needle stick injury, open wound).**
2.

3. **Chemical hazards:**
    - **a) Injury due to toxic or harmful chemicals (Potassium cyanide, formalin).**
    - **b) Fire due to flammable chemicals (Acetone, Alcohol, xylene).**
    - **c) Burn due to corrosive chemicals (HCl, H_2SO_4, NaOH, KOH, etc.).**
4.
5. **Fire hazard: Burn, injury.**
6. **Electrical hazard**
7. **Physical hazards**
8. **Hazard due to broken glass wares.**

Biohazard: Any infectious materials donated by the patient that may risk the health of laboratory personnel are called biohazards.

Types of biohazardous waste:

1. **Human blood and blood products**
2. **Human body fluid**
3. **Microbiological waste**
4. **Pathological waste**
5. **Animal waste**
6. **Sharp waste**

Route of entry of biohazard:

1. **Direct contact with broken skin**
2. **During mouth pipetting**
3. **Accidental aspiration**
4. **Accidental inoculation**
5. **Accidental inhalation**

Management of biohazards:

1. **Waste collection**
2. **Segregation**
3. **Transportation and storage**
4. **Treatment and disposal**
5. **Disposal**

Treatment of disposal biohazards:

1. **Incineration**
2. **Autoclaving**
3. **Chemical method**
4. **Plasma pyrolysis**

Q32. What is quality control of a laboratory and how can it be maintained?

Laboratory quality control: It is designed to detect, reduce, and correct deficiencies in a laboratory's internal analytical process prior to the release of patient results, in order to improve the quality of the results reported by the laboratory.

Types of quality control:

1. **External quality control:**
    - **a) Local**
    - **b) National**
    - **c) International**
2.
3. **Internal quality control:**
    - **a) Preventive**
    - **b) Corrective**
4.

Important reliable criteria of QC:

1. **Accuracy**
2. **Precision**
3. **Specificity**
4. **Sensitivity**

Benefit of QC:

1. **Immediate information is found.**
2. **The quality of lab work improves day by day.**
3. **Saves time and money.**
4. **The proper method can be identified.**
5. **Early recognition of reagent & equipment damage.**

How can we maintain QC of the lab?

1. **Use of quality equipment, reagents, and logistics.**
2. **Employment of educated and experienced lab personnel.**
3. **Maintenance of proper analytical skill throughout the test.**
4. **Solution of an accurate and precise method.**
5. **Check interpretation before reporting.**
6. **Check the report before dispatch to the patient.**
7. **Use of standard solutions:**
    - **Reagent**
    - **Reference**
    - **Specimen**
    - **Statistical methods to evaluate the obtained results.**

Q33. What are the differences between serum and plasma? Name some biochemical tests which are done by fresh plasma. What are the emergency tests that we have done in the laboratory?

Biochemical tests which are done by fresh plasma:

1. Prothrombin time (PT)
2. Activated partial thromboplastin time (APTT)
3. D-dimer
4. Fibrin degradation product
5. Thrombin time
6. Other coagulation factor tests.

Emergency tests:

1. Blood glucose
2. Blood urea
3. Serum creatinine
4. Serum electrolytes
5. Serum calcium
6. Serum phosphate
7. Arterial blood gas analysis (ABG)
8. Serum potassium
9. Serum ammonia

Differences between Plasma and Serum:

Plasma	**Serum**
Fluid obtained when anticoagulated blood has been centrifuged.	Fluid obtained when coagulated blood has been centrifuged.
An anticoagulant is needed for separation.	Anticoagulants are not needed.
Fibrinogen is present in plasma.	Fibrinogen is absent.
Does not need standing; it could be centrifuged as soon as it has been mixed thoroughly.	Serum takes a longer time to prepare.
Plasma is delivered to patients who lack blood cells.	Serum is the most preferred part of the blood used in checking blood groups and diagnosis of disease.

Q34. Name the common analytical processes used in the clinical lab. Write the principle of a photometer. What are the differences between photometry and electrophoresis?

Name of the common analytical processes used in the clinical lab:

1. **Chromatography**
2. **Electrophoresis**
3. **Photometry (Colorimeter and Spectrophotometer)**
4. **Fluorimetry**
5. **Flame photometry**
6. **Ultracentrifugation**
7. **Radioimmunoassay (RIA)**
8. **ELISA**
9. **Auto-analyzer**
10. **Enzyme immunoassay (Homogenous and heterogeneous)**

Principle of photometer: When a light at a particular wavelength is passed through a solution, some amount of it is absorbed, and therefore the light that comes out is diminished. The nature of light absorption in a solution is governed by the Beer-Lambert law.

Differences between Electrophoresis and Photometry:

Electrophoresis	**Photometry**
The movement of a charged particle in an external electric field toward the oppositely charged electrode is known as electrophoresis.	A photometer is an instrument used to quantify different parts of light.
Types: Boundary, Zone, Immuno electrophoresis.	Types: Spectrophotometer, Colorimeter.
It is used for the separation of enzymes, antigens, etc.	It is used to determine the amount of an analyte in a solution.

It separates charged particles.	It does not separate charged particles.
It does not follow the Beer-Lambert law.	It follows the Beer-Lambert law.

Q35. Name the four major components of gastric juice with each function. Why is normal gastric juice secretion essential for erythropoiesis? Name the protein-splitting enzymes of the GIT.

Components of gastric juice:

1. **Pepsin:**
    - **Function: Pepsin acts on protein to break it down to the stage of peptone.**
2.
3. **HCl (Hydrochloric Acid):**
    - **Function:**
        - **a) Converts inactive pepsinogen into active pepsin.**
        - **b) Helps in the digestion of protein and the absorption of iron.**
        - **c) Kills many ingested bacteria.**
    -
4.
5. **Intrinsic factor of Castle:**
    - **Function: It binds with vitamin B_{12} and is necessary for its absorption from the small intestine.**
6.
7. **Mucus:**
    - **Function:**
        - **a) Mucus is responsible for protecting the gastric mucosa from the aggressive action of HCl. Thus, the stomach does not self-digest.**
        - **b) Prevents PUD (Peptic ulcer disease).**
    -
8.

Name the protein-splitting enzymes of the GIT:

1. **Pepsin**
2. **Trypsin**
3. **Chymotrypsin**
4. **Elastase**

5. **Aminopeptidase**
6. **Carboxypeptidase**

Why is gastric juice secretion essential for erythropoiesis?
The extrinsic hematopoietic factor, Vitamin B_{12}, is essential for the maturation of RBCs, and its deficiency leads to pernicious anemia. One of the constituents of gastric juice, the intrinsic factor of Castle, helps in the absorption of vitamin B_{12}, which is released from the pyloric mucosa. Gastric mucosal atrophy, partial or total gastrectomy, or gastrojejunostomy may lead to decreased gastric intrinsic factor secretion, which causes pernicious anemia. That's why gastric juice secretion is essential for erythropoiesis.

Q36. What are the biochemical changes occurring in collected blood when examination is delayed? How is hemolysis of blood prevented? Write down the preparation to be taken during sample collection?

Biochemical changes occur:

1. **Diffusion of K^+ from cell → high K^+ in serum/plasma.**
2. **Conversion of pyruvate and glucose into lactic acid → high lactic acid.**
3. **Loss of CO_2.**
4. **High Cl^- in the cell due to the chloride shift.**
5. **High plasma inorganic phosphate.**
6. **Increased ammonia concentration.**
7. **Decrease in RBC count, pH, protein factor, Na^+, DPG.**
8. **Decrease in clotting factors.**
9. **Inactivation of the Na^+-K^+ pump.**

Prevention of hemolysis:

1. **Blood should be collected with moderate suction.**
2. **The needle should be removed before transferring blood into vials.**
3. **Avoid the use of wet and dirty tubes.**
4. **Avoid shaking of whole blood before clotting.**
5. **Avoid freezing and thawing of the blood specimen.**
6. **Blood should be drawn into the syringe with a gentle pull.**
7. **Excessive anticoagulant should be avoided.**

Precaution during sample collection:

1. **Identify the proper patient and check the test list before sample collection.**
2. **Label the tube with the patient ID and name before sample collection.**
3. **The tube must be sterilized, clean, and dry properly.**
4. **A disposable syringe should be used.**

5. The tourniquet should be applied tightly but not for a prolonged time.
6. The area of the prick should be dried and cleaned.
7. Separate serum/plasma after proper clotting/centrifugation.
8. Whole blood should not be frozen for a prolonged time.
9. Cells and serum should be separated before preservation.

Q37. Write down a normal daily water intake and output chart.

Condition	Nature of Intake / Output	Obligatory (ml/day)	Facultative (ml/day)
Intake	Drinks (pure water & beverage)	650	1000
	Food water	600	
	Metabolic water	350	
	Total →	1600-2600 ml/day	
Output	As urine	500	1000
	Perspiration (via skin)	550	
	Transpiration (via expiration)	400	
	With stool	150	

Total →	**1600-2600 ml/day**

Q38. Define recombinant DNA technology. Write down the application of recombinant DNA technology.

Recombinant DNA technology: The genetic manipulation in the laboratory involving the isolation and cloning of specific DNA is collectively referred to as recombinant DNA technology.

Application:

1. **Understand a number of diseases on a molecular basis. e.g., Sickle cell anemia.**
2. **Production of large amounts of protein for therapeutic purposes. e.g., recombinant insulin, GH.**
3. **Production of protein for vaccines. e.g., Hep-B vaccine.**
4. **Application to forensic medicine, e.g., to identify criminals and parents.**
5. **Gene therapy for sickle cell anemia and thalassemia.**
6. **Prenatal diagnosis of genetic diseases.**
7. **Tissue typing for organ transplantation.**

Q39. Short note on PCR.

Definition: Cloning (amplification) of a specific DNA molecule to make a large number of identical copies by an artificial, in vitro, cell-free amplification process.

Requirements:

1. **Target DNA to be cloned: It acts as a template DNA in PCR.**
2. **Taq DNA polymerase.**
3. **A pair of primers - Forward and reverse.**
4. **All four deoxy-nucleoside triphosphates (dNTPs): e.g., dATP, dGTP, dCTP, dTTP.**

Steps of PCR:

1. **Isolation of the target DNA molecule to be cloned.**
2. **Construction of two primers.**
3. **Mixing of target DNA with other accessories.**

4. **Denaturation of DNA: Heating the mixture by raising the temperature up to 95-100°C for 30 sec to 60 sec. This causes the denaturation of target dsDNA into two ssDNA strands.**
5. **Annealing of primers: Cooling of the mixture by lowering the temperature to 45-54°C for 30 sec to 60 sec. This causes the annealing of primers with ssDNA at their flanking regions.**
6. **Extension of primer / DNA amplification: Heating the mixture by raising the temperature to 72°C for 30-60 sec. This causes elongation of primers and synthesis of new complementary DNA by polymerization of the added dNTPs, catalyzed by Taq DNA polymerase.**
7. **Steps 4, 5, and 6 constitute one cycle of PCR, and it is repeated again and again.**

Importance / Application of PCR:

1. **Diagnosis of infectious diseases, e.g., bacterial and viral.**
2. **Production of hormones, proteins, and vaccines.**
3. **Medicolegal use for criminal identification and paternity testing.**
4. **DNA analysis and diagnosis of genetic diseases by detecting mutations.**
5. **Cancer detection by detecting mutations in oncogenes.**
6. **Tissue typing for transplantation surgery.**
7. **Prenatal diagnosis of genetic diseases.**

Q40. Short note on thyroid function tests.

Thyroid function tests:

1. **Measurement of plasma TSH concentration:**
    - **High TSH → Primary hypothyroidism.**
    - **Low TSH → Secondary hypothyroidism.**
    - **Very low/undetectable → Hyperthyroidism.**
2.
3. **Measurement of plasma thyroid hormone concentration:**
    - **a) T3 (Triiodothyronine)**
    - **b) T4 (Thyroxine)**
    - **Decreases in T3 and T4 → Hypothyroid.**
    - **Increases in T3 and T4 → Hyperthyroid.**
    - **Free T3 (FT3) and Free T4 (FT4) are preferably measured.**
4.
5. **Radioactive iodine uptake test (RAUT)**
6. **Measurement of thyroid autoantibody**
7. **TRH test**
8. **TSH stimulation test**
9. **T3 suppression test**

Q41. Name the thyroid diseases and tests related to them.

Thyroid diseases and tests:

1. **Hypothyroidism: It is the clinical condition produced by insufficient circulating thyroid hormones due to their decreased synthesis.**
    - **Lab tests:**
        - **a) TSH → High**
        - **b) T3 → Low**
        - **c) T4 → Low**
        - **d) Radioactive iodine uptake → low uptake**
        - **e) Measurement of thyroid antibodies → Positive in autoimmune thyroiditis.**
    -
2.
3. **Hyperthyroidism: It is the clinical condition produced by excess circulating thyroid hormones due to their increased synthesis.**
    - **Laboratory tests:**
        - **a) TSH → Low**
        - **b) T3/FT3 → High**
        - **c) T4/FT4 → High**
        - **d) Radioactive iodine uptake test → Increase**
        - **e) Thyroid autoantibodies → Positive**
        - **f) T3 Suppression test → Fail**
    -
4.
5. **Goiter: Enlargement of the thyroid gland.**
    - **Lab tests:**
        - **a) S. iron → Low**
        - **b) T3 and T4 → Low**
        - **c) TSH → High**
        - **d) Thyroglobulin → High**
    -
6.

Q42. Justify that bile is a digestive juice. Describe the steps of carbohydrate digestion and absorption.

Justification of bile as a digestive juice: Bile is a digestive juice, though it has no digestive enzymes.

Functions of bile:

1. **Bile salts in bile help in fat digestion by emulsification of fat.**
2. **Helps in the absorption of fat and other lipids.**
3. **Bile serves as a medium for the excretion of bile pigment (bilirubin), cholesterol, drugs, etc.**

Common digestive juices:

Name	Source	pH
1. Saliva	Salivary gland	6.7-7.0
2. Gastric juice	Gastric gland	1.5-2.0
3. Pancreatic juice	Pancreas	8.0-8.3
4. Bile	Liver	7.8
5. Succus entericus	Small intestinal mucosa	7.5

Digestion and absorption of carbohydrates (CHO):

- **In the mouth: Digestion is started by salivary α-amylase. Starch is converted to maltose and maltotriose.**
- **In the stomach: No digestion occurs in the stomach.**
- **In the small intestine: Salivary α-amylase and pancreatic α-amylase act on starch. 20%-40% of digestion occurs due to α-amylase, and 50-80% of digestion occurs due to pancreatic amylase.**
- **In the brush border:**
    - **Lactase: Lactose → Glucose + Galactose**
    - **Maltase: Maltose → Maltotriose + Glucose**
    - **Sucrase: Sucrose → Glucose + Fructose**
-

Absorption of CHO: CHOs are mainly absorbed as fructose, galactose, and glucose. Glucose and galactose are reabsorbed by secondary active transport (Na^+-glucose co-transport). Fructose is absorbed by facilitated diffusion from the intestinal lumen into the enterocyte.

Q43. Write a short note on Anticoagulants.

Definition: Anticoagulants are chemical substances which prevent the coagulation of blood by interfering with the coagulation mechanism through their chemical action.

Classification (According to the mode of action):

1. **Acting in vitro: Citrate, oxalate, EDTA, heparin, etc.**
2. **Acting in vivo: Warfarin, phenindione.**
3. **Both in vitro and in vivo: Heparin, dextran sulphate.**

Other classification:

1. **Natural anticoagulant: Heparin, antithrombin-III, fibrin.**
2. **Artificial anticoagulant: Heparin, Sodium citrate, EDTA, Double oxalate (Paul-Heller mixture), Sodium fluoride, Warfarin, Citrate phosphate dextrose.**

Q44. Short note on SI units.

Definition: The newest metric measurement system is called "System International" (SI units). It is considered the best metric measurement system. It was revised by the general conference on weight and measures held in Paris in the early 1960s.

Basic SI units:

Physical quantity	Name of SI unit	Symbol
Length	Meter	m
Mass/Weight	Kilogram	kg

Time	**Second**	**s**
Electric current	**Ampere**	**A**
Thermodynamic temperature	**Kelvin**	**K**
Amount of substance	**Mole**	**mol**

Importance of SI units: Its terms and symbols are standard in all languages and disciplines. Therefore, SI promotes better communication and understanding throughout the world, both within the biological and medical communities and among the various scientific and engineering disciplines.

Q45. Short note on serum electrolytes.

Electrolytes: Electrolytes are molecules that dissociate in water into their cation and anion equivalents and are capable of carrying an electric charge.

- **Important electrolytes: Na^+, K^+, Cl^-, HCO_3^-.**
- **Clinically, these 4 electrolytes are measured frequently and routinely in the name of an "electrolyte profile" for patient management.**

Major electrolyte disorders:

1. **Disorder of sodium balance:**
    - **a) Hyponatremia: Decrease in Na^+**
    - **b) Hypernatremia: Increase in Na^+**
2.
3. **Disorder of potassium balance:**
    - **a) Hypokalemia: Decrease in K^+**
    - **b) Hyperkalemia: Increase in K^+**
4.
5. **Disorders of calcium balance:**
    - **a) Hypocalcemia: Decrease in Ca^{2+}**
    - **b) Hypercalcemia: Increase in Ca^{2+}**
6.

Q46. Short note on cardiac markers.

Definition: These are intracellular enzymes or proteins released into the blood following cardiac muscle damage due to ischemia. There is a serial change in plasma concentration with an initial rise followed by a fall.

Indication:

1. **Diagnosis: To diagnose Myocardial Infarction (MI).**
2. **Prognosis: To assess the prognosis of MI.**
3. **Risk stratification: Risk stratification of disease involving MI.**

Types of cardiac markers: There are two types.

1. **Enzymes:**
    - **Creatine kinase (CK) or creatine phosphokinase (CPK)**
    - **Aspartate transaminase (AST)**
    - **Lactate dehydrogenase**
    - **CK-MB**
2.
3. **Proteins:**
    - **Troponins:**
        - **Troponin-I**
        - **Troponin-T**
        - **Troponin-C**
    -
    - **Myoglobin**